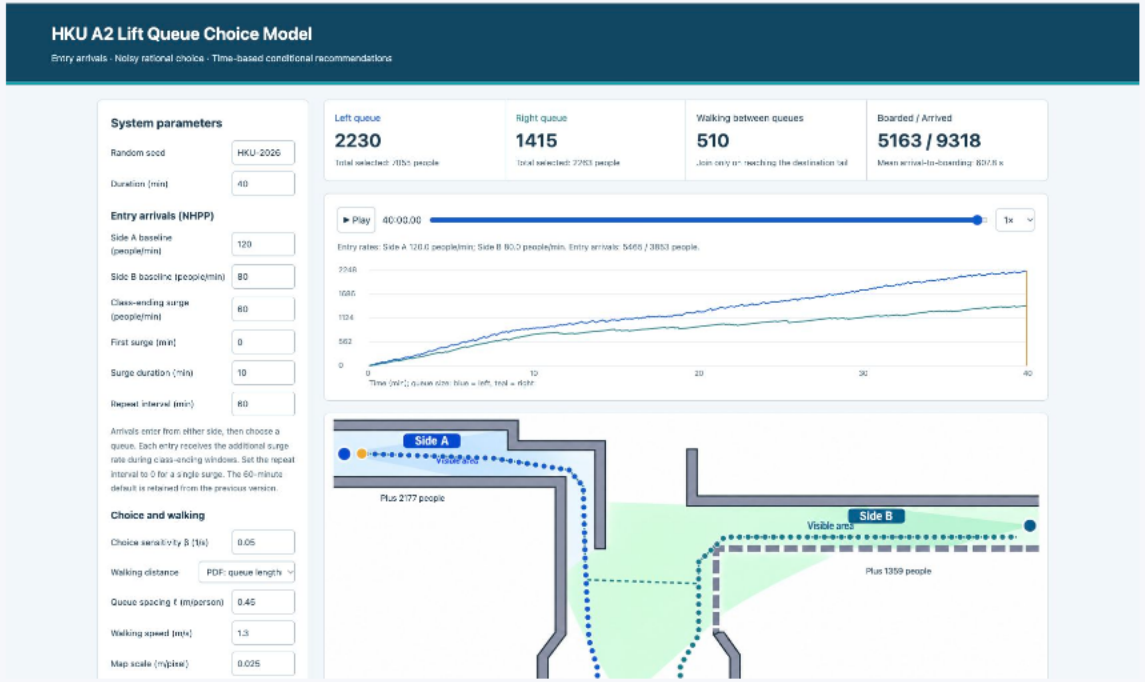
ACADEMIC SIMULATION WEBSITE



Public website: liyufei2004.github.io/hku-a2-queue-simulation/

Parameter controls
Arrival rates, class surges, lift cycles and walking assumptions

Inspectable replay
See queues, walkers, lift events and the choice-cost breakdown

Reproducible exports
Seeded runs; parameter-tagged CSV / JSON

CONDITIONAL DECISION SUPPORT

Entry + time window + local queue length

Estimate the hidden queue distribution
Compare conditional expected delays
Report support, or withhold a recommendation

1,000,000 paired snapshots
2,084 seeded trajectories · 5-second sampling · 5-minute bins

Fewer than 30 matches: no definite recommendation. Snapshots within a run are correlated.

EXPERIMENT DESIGN

Four scenarios, 30 independent runs each

Baseline inflow
70 or 200 people per minute

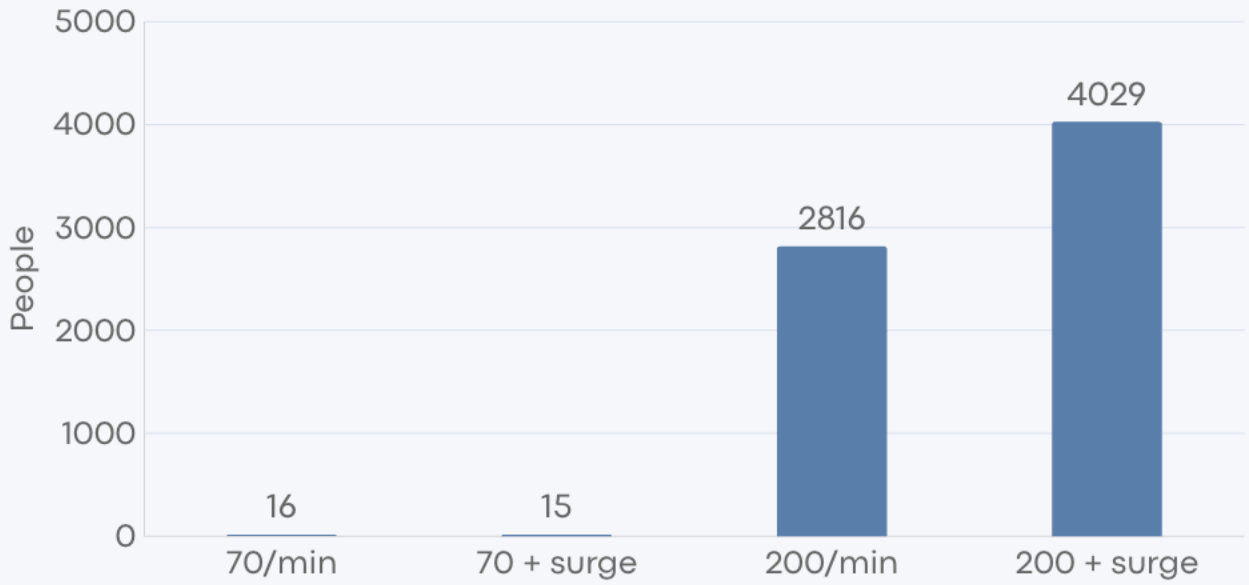
Class-ending surge
None, or +60/min at each entry for minutes 0–10

Fixed assumptions
40-minute horizon; four lifts per bank
Mean cycles: 40 s / 200 s
Capacity limit: 20; mean trip capacity: 18

Nominal total service: 129.6 people/min

Illustrative scenarios, not calibrated field estimates. API/model checks and passenger conservation passed.

DEMAND AND CAPACITY



4,029 ± 42 people not boarded at 40 min
(default + surge)

70/min can clear the surge backlog; 200/min cannot clear it within this horizon.

Mean across 30 runs; ±95% t interval. Includes queued + walking passengers. No claim of real-world delay savings.